



Best Paper Award United States 2026

Celebrating the Award Recipients

A Real-time Prognostic-Based Control Framework for Hybrid Electric Vehicles

Laxman Timilsina, Phuong H. Hoang, Ali Moghassemi, Elutunji Buraimoh, Phani Kumar Chamarthi, Gokhan Ozkan, Behnaz Papari, and Christopher S. Edrington

Clemson University ▪ Operation Technology Inc.



**Best
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GaN Four-leg Inverter Implementing Novel Common Mode Elimination using a Hardware-in-the-Loop System-Level Controller

Caleb Li, Annette von Jouanne, Giovanna Oriti,
Alex Julian, Emmanuel Agamloh, and Alex Yokochi

Baylor University ▪ Naval Postgraduate School ▪ Seaside



Best Paper Award



Advancing Power System Time Synchronization: A PTP-Based Solution with Hardware-in-the-Loop

Yifu Li, Abouzar Estebarsari, Richard Hoptroff,
Payman Dehghanian, and Mohammad Alhazmi

George Washington University ▪ London South Bank University ▪ Hoptroff Ltd. ▪ King Saud University



**Best
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Query-and-Response Digital Twin Framework using a Multidomain, Multifunction Image Folio

Kerry Sado, Jarrett Peskar, Austin R. J. Downey,
Herbert L. Ginn, Roger Dougal, and Kristen Booth

University of South Carolina



Outstanding Paper Award



Optimal Design and Modeling of a Hybrid Energy Storage System Including Battery and Hydrogen in DC Microgrids

Massiagbe Fatoumata Diabate, Harish Sarma Krishnamoorthy, and Jian Shi

University of Houston



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Development, Integration, and Flight Testing of a Silicon Carbide Propulsion Drive for a Hybrid Electric Aerospace Application

Chris Farnell, Anna Corbitt, Wesley G. Schwartz, Asif Faruque, Yue Zhao, David Huitink, Nenad Miljkovic, and H. Alan Mantooth

University of Arkansas ▪ University of Illinois at Urbana-Champaign



Distinguished Paper Award



Hierarchical Partial Power Processing of Second-Life Battery-Integrated Grid-Connected Multilevel Inverters via Statistical Design

Rupeng Duan, Xiao Sun, and Xiaofan Cui

University of California, Los Angeles



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A Novel Digital Twin for Diesel Generator in Microgrids

Renuka Loka, Mahmoud Kabalan, Thomas Holliday, Rakesh Shamrao Patekar, Mohanad Elsayed, Mohamed M. Zakaria Moustafa, and Thomas Bozada

Center for Microgrid Research - University of Saint Thomas ▪ US Army Engineer Research and Development Center - Construction Engineering Research Laboratory



Distinguished Paper Award





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